

Vinod Raman

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<https://vinodkraman.github.io>

Education

University of Michigan

PhD Statistics

Thesis Advisor: Ambuj Tewari

Ann Arbor, MI

Sep. 2021 - Dec. 2025

University of Michigan

BSE Computer Science, BSE Chemical Engineering

Thesis Advisors: Mahdi Cheraghchi, Sindhu Kuttly, Andrej Lenert

Ann Arbor, MI

Sep. 2015 - May 2020

Experience

Google DeepMind

Research Scientist

New York, NY

Dec. 2025 - Present

- Improving Gemini's fundamental capabilities for generative and agentic retrieval.
- Using Reinforcement Learning to improve the quality and diversity of synthetic data.
- Improving reasoning efficiency and diversity of LLMs via latent thinking and reward shaping.

Google Research

Research Intern, Student Researcher

Hosts: Matthew Joseph, Travis Dick, Umar Syed

New York, NY

June - Nov. 2025

- Designed and implemented new state-of-the-art private algorithms for domain discovery, top- k selection, and submodular maximization over unbounded domains. Paper accepted as an Oral at ICLR 2026.
- Developed an evaluation framework for arithmetic reasoning in frontier LLMs. Paper in submission.
- Built a scalable synthetic data generation pipeline for LLM training, combining fine-tuning and filtering to produce high-quality data at scale.

Apple

Research Intern

Host: Satyen Kale

New York, NY

Feb. - June 2025

- Designed and implemented adaptive allocation strategy for Best-of- N alignment for batch LLM inference.
- Proposed efficient methods for sample-dependent mass estimation. Paper in progress.

Apple

Research Intern

Host: Kunal Talwar

Cupertino, CA

May - Aug. 2024

- Developed state-of-the-art differentially private algorithms for adversarial bandits and online learning for dynamic regret minimization.
- Research published in two papers at ICML 2025.

Awards & Fellowships

- Outstanding Dissertation Award, 2026
- ALT Outstanding Paper Award, 2025
- Apple Scholars in AI/ML PhD Fellowship, 2025
- Outstanding First-Year Ph.D. Student, 2022
- NSF Graduate Research Fellowship, 2022

Publications

(*) denotes equal contribution. (α - β) denotes alphabetical ordering

1. (α - β) T. Dick, M. Joseph, **V. Raman**. Missing Mass for Differentially Private Domain Discovery. *International Conference on Learning Representations (ICLR)*, 2026. **Oral Presentation**.
<https://arxiv.org/abs/2603.14016>
2. S. Somerstep, **V. Raman***, U. Subedi*, Y. Sun. Learning to Choose or Choosing to Learn: Best-of-N vs. Supervised Fine-Tuning for Bit String Generation. *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2026.
<https://www.arxiv.org/abs/2505.17288>
3. J. Li*, **V. Raman***, A. Tewari. Generation through the lens of learning theory. *Conference on Learning Theory (COLT)*, 2025.
<https://arxiv.org/abs/2410.13714>
4. A. Raman*, **V. Raman***. Generation from Noisy Examples. *International Conference on Machine Learning (ICML)*, 2025.
<https://arxiv.org/abs/2501.04179>
5. C. Peale*, **V. Raman***, O. Reingold*. Representative Language Generation. *International Conference on Machine Learning (ICML)*, 2025.
<https://arxiv.org/abs/2505.21819>
6. H. Asi*, **V. Raman***, A. Saha*. Tracking the Best Expert Privately. *International Conference on Machine Learning (ICML)*, 2025.
<https://arxiv.org/abs/2503.09889>
7. **V. Raman**, H. Asi, K. Talwar. Faster Rates for Private Adversarial Bandits. *International Conference on Machine Learning (ICML)*, 2025.
<https://arxiv.org/abs/2505.21790>
8. **V. Raman***, U.Subedi*, A.Tewari. The Complexity of Sequential Prediction in Dynamical Systems. *Conference on Learning for Dynamics and Control (L4DC)*, 2025. **Oral Presentation**.
<https://arxiv.org/abs/2402.06614>
9. **V. Raman***, U.Subedi*, A.Tewari. A Unified Theory of Supervised Online Learnability. *Conference on Algorithmic Learning Theory (ALT)*, 2025. **Outstanding Paper Award**.
<https://arxiv.org/abs/2307.03816>
10. **V. Raman***, U.Subedi*, A.Tewari. A Characterization of Multioutput Learnability. *Journal of Machine Learning Research (JMLR)*, 2024.
<https://arxiv.org/abs/2301.02729>
11. **V. Raman**, A.Tewari. Online Classification with Predictions. *Conference on Neural Information Processing Systems (NeurIPS)*, 2024.
<https://arxiv.org/abs/2405.14066>
12. S.Hanneke*, **V. Raman***, A. Shaeiri*, U.Subedi*. Multiclass Transductive Online Learning. *Conference on Neural Information Processing Systems (NeurIPS)*, 2024. **Spotlight**.
13. **V. Raman***, U.Subedi*, A.Tewari. Smoothed Online Classification can be Harder than Batch Classification. *Conference on Neural Information Processing Systems (NeurIPS)*, 2024.
<https://arxiv.org/pdf/2405.15424>

14. **V. Raman***, U.Subedi*, A. Raman, A.Tewari. Apple Tasting: Combinatorial Dimensions and Minimax Rates.
Conference on Learning Theory (COLT), 2024.
<https://arxiv.org/abs/2310.19064>
15. **V. Raman***, U.Subedi*, A.Tewari. Online Learning with Set-Valued Feedback.
Conference on Learning Theory (COLT), 2024.
<https://arxiv.org/abs/2306.06247>
16. **V. Raman***, U.Subedi*, A.Tewari. Online Infinite-Dimensional Regression: Learning Linear Operators.
Conference on Algorithmic Learning Theory (ALT) 2024.
<https://arxiv.org/abs/2309.06548>
17. A.Raman, **V. Raman***, U.Subedi*, I.Mehalel*, A.Tewari. Multiclass Online Learnability under Bandit Feedback.
Conference on Algorithmic Learning Theory (ALT) 2024.
<https://arxiv.org/abs/2308.04620>
18. **V. Raman***, U.Subedi*, A.Tewari. On Proper Learnability between Average- and Worst-case Robustness.
Conference on Neural Information Processing Systems (NeurIPS), 2023.
<https://arxiv.org/abs/2211.05656>
19. **V. Raman***, U.Subedi*, A.Tewari. On the Learnability of Multilabel Ranking.
Conference on Neural Information Processing Systems (NeurIPS), 2023. **Spotlight**.
<https://arxiv.org/abs/2304.03337>
20. (α - β) S.Hanneke, S.Moran, **V. Raman**, U.Subedi, A.Tewari. Multiclass Online Learning and Uniform Convergence.
Conference on Learning Theory (COLT), 2023.
<https://arxiv.org/abs/2303.17716>
21. **V. Raman**, A.Tewari. Online Agnostic Multiclass Boosting.
Conference on Neural Information Processing Systems (NeurIPS), 2022.
<https://arxiv.org/abs/2205.15113>
22. **V. Raman**, T.Burger, A.Lenert. Design of thermophotovoltaics for tolerance of parasitic absorption.
Optics Express, 27(22):31757–31772, 2019.
<https://doi.org/10.1364/OE.27.031757>

In Submission

1. J. Leng, S. Si, **V. Raman**, H. Yu, I. Dhillon. GroupDPO: Memory efficient Group-wise Direct Preference Optimization.
In Submission, 2026.
<https://arxiv.org/abs/2604.15602>
2. J. Li, **V. Raman**, A. Tewari. Generation in Metric Spaces.
In Submission, 2026.
<https://arxiv.org/abs/2602.07710>
3. Y. Kalayci*, **V. Raman***, S. Dughmi. Optimal Stopping vs. Best-of- N for Inference-time Optimization.
In Submission, 2026.
<https://arxiv.org/abs/2510.01394>

Preprints

1. $(\alpha-\beta)$ V. Feldman, S. Kale, **V. Raman**, K. Talwar, A. Tewari. Estimating the (Un)seen: Sample-dependent Mass Estimation. *Preprint*, 2025.
2. S. Xie, **V. Raman**, S. Zhou. Transductive and Learning-Augmented Online Regression. *Preprint*, 2025.
3. **V. Raman**, H. Asi, S. Kale. AdaBoN: Adaptive Best-of-N Alignment. *Preprint*, 2025.
<https://arxiv.org/abs/2505.12050>
4. **V. Raman***, D.Zhang*, Y.Jung, A.Tewari. Online Boosting for Multilabel Ranking with Top- k Feedback. *Preprint*, 2020.
<https://arxiv.org/abs/1910.10937>

Selected Talks

1. Optimal Stopping vs. Best-of- N for Inference-time Optimization. *Google DeepMind Tech Talk*, 2025.
2. A Unified Theory of Supervised Online Learning. *ALT*, 2025.
3. Generation through the lens of learning theory. *Apple MLR Reading Group*, 2025.
4. Generation through the lens of learning theory. *NEU CS Theory Seminar*, 2024.
5. Trichotomies in Online Learnability. *Apple MLR Reading Group*, 2024.
6. Multiclass Online Learning and Uniform Convergence. *UMich EECS Theory Seminar*, 2024.
7. On Classification-Calibration of Gamma-Phi Losses. *COLT*, 2023.

References

1. **Ambuj Tewari**, Professor, Statistics, University of Michigan, Ann Arbor MI, USA.
Email: tewaria@umich.edu | *Phone*: 734-615-0928
2. **Steve Hanneke**, Assistant Professor, Computer Science, Purdue University, West Lafayette IN, USA.
Email: steve.hanneke@gmail.com
3. **Mahdi Cheraghchi**, Associate Professor, Computer Science, University of Michigan, Ann Arbor MI, USA.
Email: mahdich@umich.edu | *Phone*: 734-763-9165